1. $(A\ I)=\begin{pmatrix} a_1 & & & 1 & & \\ & a_2 & & & 1 & \\ & & \ddots & & & \ddots \\ & & & a_n & & & 1 \end{pmatrix} \to \begin{pmatrix} 1 & & & a_1^{-1} & & \\ & 1 & & & a_2^{-1} & \\ & & \ddots & & & \ddots \\ & & & 1 & & & a_n^{-1} \end{pmatrix}$

$\therefore A^{-1}=\begin{pmatrix} a_1^{-1} & & \\ & a_2^{-1} & \\ & & \ddots \\ & & & a_n^{-1} \end{pmatrix}$

2. 设 $A=\begin{pmatrix} 1 & -1 & 3 \\ 2 & -1 & 4 \\ -1 & 2 & -4 \end{pmatrix}$ $(A\ I)=\begin{pmatrix} 1 & -1 & 3 & 1 & 0 & 0 \\ 2 & -1 & 4 & 0 & 1 & 0 \\ -1 & 2 & -4 & 0 & 0 & 1 \end{pmatrix} \to \begin{pmatrix} 0 & 1 & -1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ -1 & 2 & -4 & 0 & 0 & 1 \end{pmatrix}$

$\to \begin{pmatrix} 0 & 1 & -1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & -3 & 1 & -1 \end{pmatrix} \to \begin{pmatrix} 0 & 1 & 0 & 4 & -1 & 2 \\ 1 & 0 & 0 & -4 & 2 & -1 \\ 0 & 0 & 1 & -3 & 1 & -1 \end{pmatrix} \to \begin{pmatrix} 1 & & & -4 & 2 & -1 \\ & 1 & & 4 & -1 & 2 \\ & & 1 & -3 & 1 & -1 \end{pmatrix}$

$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=A^{-1}\begin{pmatrix} 8 \\ 11 \\ -11 \end{pmatrix}=\begin{pmatrix} -4 & 2 & -1 \\ 4 & -1 & 2 \\ 3 & -1 & 1 \end{pmatrix}\begin{pmatrix} 8 \\ 11 \\ -11 \end{pmatrix}=\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$

4. $(A^k)^{-1}\cdot A^k=I=(A^{-1})^k A^k=(A^{-1}\cdots A^{-1})(A\cdots A)=(A^{-1})^{k-1}A^{-1}A(A)^{k-1}$

易知 A 可逆，A 与 A^k 可交换 $\therefore (A^k)^{-1}=(A^{-1})^k$

6. $\because I_n+A$ 是非异阵 $\therefore |I_n+A|\neq 0$ 又 $\because A^2=I_n$ $\therefore (I_n+A)(I_n-A)=0$

$\therefore I_n-A=0$ $I_n=A$

7. $\because A^k=A^t$ ~~$\therefore |A_{ij}|$ 的代数余子式 $=a_{ij}$~~ $\therefore A_{ij}=a_{ij}$

以第 j 行展开 $\therefore |A|=a_{j1}^2+a_{j2}^2+\cdots+a_{jn}^2=\sum_{i=1}^{n} a_{jn}^2$

又 $\because A$ 非零 故 $\exists\ a_{ij}\neq 0$ $\therefore |A|\neq 0$

$\therefore A$ 是非异阵

8. $A+B=A(I+A^{-1}B)=A(B^{-1}+A^{-1})B$

$\therefore A(B^{-1}+A^{-1})B(A+B)^{-1}=I$ $\therefore (B^{-1}+A^{-1})B(A+B)^{-1}A^{-1}=A^{-1}A=I$

$\therefore A^{-1}+B^{-1}$ 也是为非异阵

9. $\because I_n-A^m=I_n$ $\therefore (I_n-A)(I_n+A+A^2+\cdots+A^{m-1})=I_n$

$\therefore I_n-A$ 是非异阵

10. $\because A^2=A$ $\therefore A^2-A-2I=-2I=(A-2I)(A+I)$

$\therefore (I-\frac{A}{2})(I+A)=I$

$\therefore A+I_n$ 是非异阵

11. $\because A^2-A-3I_n=0$ $\therefore (A-2I_n)(A+I_n)=I_n$

$\therefore A-2I_n$ 是非异阵

12. $\because A+B=AB$ $\therefore I_n=I_n-A-B+AB=(I_n-A)(I_n-B)=(I_n-B)(I_n-A)=I_n-A-B+BA$

$\therefore I_n-A$ 是可逆阵 $\therefore AB=A+B=BA$

Recording start from here

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1. (2) 原式 =

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & 0 & -2 \\ 1 & 1 & 3 & 2 \\ 2 & 2 & 6 & 4 \end{pmatrix} \xrightarrow{\substack{-1 \\ -2}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

-3

2. (1)

$$\begin{pmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 0 & 1 & 2 \\ 2 & 2 & 4 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 1 & 2 \\ 0 & -2 & -2 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

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